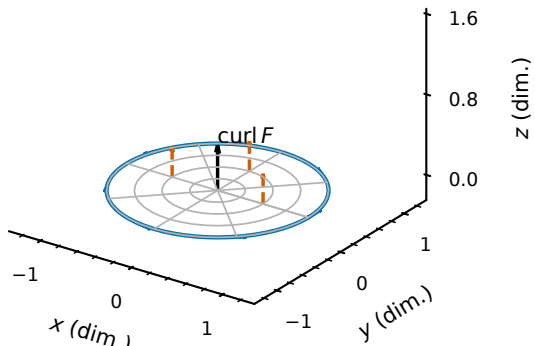
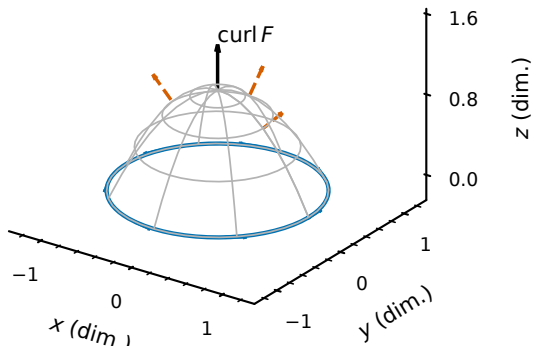
A



Disk: $\iint (\operatorname{curl} F) \cdot n \, dS = 2\pi$

B



Cap $z = 1 - r^2$: same flux 2π